

Academic Calendar for Microbiology

2 nd Phase (In months)											
1	2	3	4	5	6	7	8	9	10	11	12
General Bacteriology Parasitology Immunology			Preparation + 1st Internal Assessment	Systemic Bacteriology Virology Mycology Clinical Microbiology			Preparation + 2nd Internal Assessment	Preparatory leave	2nd Professional Exam		

ITEM CARDS

There will be 2 (two) Cards

- Item card 1:** General Bacteriology, Parasitology, Immunology
- Item card 2:** Systemic Bacteriology, Virology, Mycology and Clinical Microbiology

DEPARTMENT OF MICROBIOLOGY MEDICAL COLLEGE ITEM CARD

Batch : Tut. Batch Roll (Write in the boxes)

Student's Profile

Name:

Contact Phone No:

Address:

Guardian:

Contact Phone No:

Address:

A passport sized recent
photograph of the
student to be attached
here

GENERAL BACTERIOLOGY (First assessment Exam)

	Topic	Marks	Signature
1	Prokaryote and eukaryote, components of bacteria, cell wall of Gram positive and Gram negative bacteria, capsule, flagella, spore, classification of bacteria		
2	Growth and death of bacteria, growth requirements, classification of bacteria according to oxygen requirement, growth curve, generation time		
3	Pathogenesis of bacterial disease, exotoxin and endotoxin, Koch's postulates, their limitations, new adjuncts, molecular Koch's Postulates. Host defense against bacterial disease		
4	Sterilization, disinfection, antisepsis, different methods, their principles and uses		
5	Practical bacteriology: Use of microscope. Gram staining, Ziehl-Neelsen staining. Culture media – types, commonly used media with their use		
6	Antimicrobial drugs, their mechanism of action, resistance, selective toxicity, antibiotic combination, chemoprophylaxis, susceptibility test Bacterial genetics – plasmid, transposons, mutation, transfer of gene,		

	Multidrug resistant (MDR). Extensively drug resistant (DXR) and Pandrug resistant (PDR) bacteria.		
7	Normal flora		
8	Biosafety and biosecurity, Biomedical waste disposal		

IMMUNOLOGY (First assessment Exam)

	Topic	Marks	Signature
1	Immunity, its type, components of innate immunity, comparison between active and passive immunity, immunocompetent cells		
2	Immunogen, antigen, properties of an ideal antigen, hapten		
3	Immunoglobulin, antibody, its structure, types, function		
4	Complements, major histocompatibility complex		
5	Cytokines, mechanism of immune response, primary and secondary immune response		
6	Tolerance, hypersensitivity, autoimmune diseases		
7	Tumour immunity, transplantation, immunodeficiency		
8	Immunological reactions- basic principles and examples		

MOLECULAR BIOLOGY (First assessment)

	Topic	Marks	Signature
1	Principle of PCR, RT-PCT, Realtime PCR,		
2	Definition of DNA Cloning, DNA recombination, Genetic engineering, biotechnology, gene therapy		

MYCOLOGY (Second assessment)

	Topic	Marks	Signature
1	Basic structure of fungi, classification of fungi, antifungal drugs		
2	Superficial & cutaneous fungi- <i>Malassezia furfur</i> , dermatophytes, <i>Candida</i> .		
3	Subcutaneous, deep & opportunistic fungi- <i>Mucor</i> , <i>Rhizopus</i> , <i>Cryptococcus</i> , <i>Aspergillus</i> .		

VIROLOGY (Second assessment)

	Topic	Marks	Signature
1	Basic virology, basic structure of a virus, defective virus, prion, replication, pathogenesis of viral disease, host defense against viral infection, antiviral drugs, general scheme of lab diagnosis of viral diseases, common viral infections in Bangladesh		
2	Herpesvirus, orthomyxovirus, paramyxovirus, rubella virus		
3	Hepatitis viruses, oncogenic viruses		
4	Human immunodeficiency virus		
5	Polio virus, rabies virus, dengue virus, rotavirus, chikungunya virus, Zika virus		
6	COVID-19		

PARASITOLOGY (First assessment Exam)

	Topic	Marks	Signature
1	Basic concepts of host, parasites and their types, classification of medically important protozoa		
2	Entamoeba, free living amoeba, Giardia, Balantidium		
3	Leishmania, Trichomonas, Trypanosoma		
4	Plasmodium, Toxoplasma, Babesia		
5	Basic structure and classification of helminthes Cestode: Taenia, Echinococcus, Diphylobothrium Trematodes: Schistosoma, Fasciolopsis		
6	Nematodes: Ascaris, Enterobius, Strongyloides, Trichuris		
7	Nematodes : Hookworm, Filariasis, Onchocerca Volvulus		

SYSTEMIC BACTERIOLOGY (Second assessment)

	Topic	Marks	Signature
1	Staphylococcus		
2	Streptococcus		
3	Neisseria, causes of pyogenic meningitis		
4	Corynebacterium, Bacillus		
5	Mycobacterium		
6	Enterobacteriaceae – General properties & classification, Escherichia coli, Shigella		
7	Salmonella		
8	Vibrio, Campylobacter		
9	Pseudomonas, Proteus, Klebsiella		
10	Haemophilus, Helicobacter, Bordetella, Bacillus		
11	Anaerobic bacteria, anaerobic culture		
12	Spirochaetes, sexually transmitted disease		
13	Rickettsia, Chlamydia, Mycoplasma		

CLINICAL MICROBIOLOGY (Second Assessment Exam)

	Topics	Marks	Signature
1	Examination of stool, morphology of common parasites found in stool, diarrhea- causes and diagnosis		
2	Examination of urine, urinary tract infection- causes and diagnosis		
3	Examination of CSF, meningitis- causes and diagnosis		
4	Blood culture, pyrexia of unknown origin		
5	Examination of sputum, throat swab, pus, wound swab, pleural fluid, ascitic fluid, genital specimen. Causes of pneumonia, sore throat, wound infection, pleural effusion, ascites, vaginal discharge, urethral discharge,		
6	Basics of Hospital Acquired Infection		
7	Infection prevention and control, hand washing, donning and doffing, Preparation of disinfectants, Disposal of Medical wastes		